

DHT22 (AM2302)

Digital temperature and humidity sensor. Single-wire output. Quick-reference datasheet summary for the AirGuard prototype.

1. Key specifications

Parameter	Value
Supply voltage	3.3 V to 6 V DC (works directly with 3.3 V ESP32 logic)
Humidity range	0 to 100 % RH
Humidity accuracy	+/- 2 % RH typical (up to +/- 5 % RH worst case)
Humidity resolution	0.1 % RH
Temperature range	-40 to +80 deg C
Temperature accuracy	+/- 0.5 deg C typical
Temperature resolution	0.1 deg C
Sampling rate	Max 0.5 Hz (one reading every 2 seconds)
Output	Single-wire digital, 40-bit data frame
Current draw	About 1 to 1.5 mA while measuring, about 40 to 50 uA standby
Long-term stability	About +/- 0.5 % RH per year
Body size	About 15.1 x 25 x 7.7 mm (4-pin package)

Values above are typical figures from the manufacturer's published specifications, rounded for quick use.

2. Pinout (front view, grille facing you, pins down, left to right)

Pin	Name	Function
1	VCC	Power supply, 3.3 to 6 V
2	DATA	Single-wire data line. Needs a pull-up resistor to VCC
3	NC	Not connected
4	GND	Ground

3. Wiring to an ESP32

DHT22	ESP32	Note
VCC	3V3	Powering at 3.3 V keeps the data line safe for ESP32 GPIO
DATA	Any free GPIO (for example GPIO 4)	Add a 4.7 to 10 kOhm pull-up between DATA and VCC (many breakout boards include it)
GND	GND	Common ground

4. Data format

Each reading is 40 bits, sent most significant bit first: 16 bits humidity, 16 bits temperature, 8 bits checksum.

Field	Bits	How to read
Humidity	16	Raw value divided by 10 gives % RH

Field	Bits	How to read
Temperature	16	Raw value divided by 10 gives deg C. The top bit is a sign bit (1 means negative)
Checksum	8	Sum of the four data bytes, lowest 8 bits. Discard the reading if it does not match

5. Communication timing (approximate)

Step	Timing
Host start signal	Pull DATA low for at least 1 ms, then release high for 20 to 40 us
Sensor response	Low for about 80 us, then high for about 80 us
Bit 0	About 50 us low, then about 26 to 28 us high
Bit 1	About 50 us low, then about 70 us high
Minimum gap between reads	2 seconds

6. Practical notes for AirGuard

- Read no faster than once every 2 seconds. Faster polling returns errors or stale values.
- Keep the sensor away from the purifier exhaust and from heat sources such as the ESP32 regulator, or readings will drift.
- Always check the checksum and ignore failed reads instead of acting on them, especially before triggering an alert.
- Common Arduino and ESP32 libraries: Adafruit DHT sensor library, or DHTesp for ESP32.
- Humidity readings can be off after condensation or long exposure to very high humidity. Allow time to recover.
- The DHT22 does not measure PM2.5. Particulate readings come from the separate dust sensor.

This is a summary written for quick reference, not a copy of the manufacturer datasheet (Aosong / ASAIR). For the official document, search for the AM2302 datasheet from the manufacturer or your component supplier.